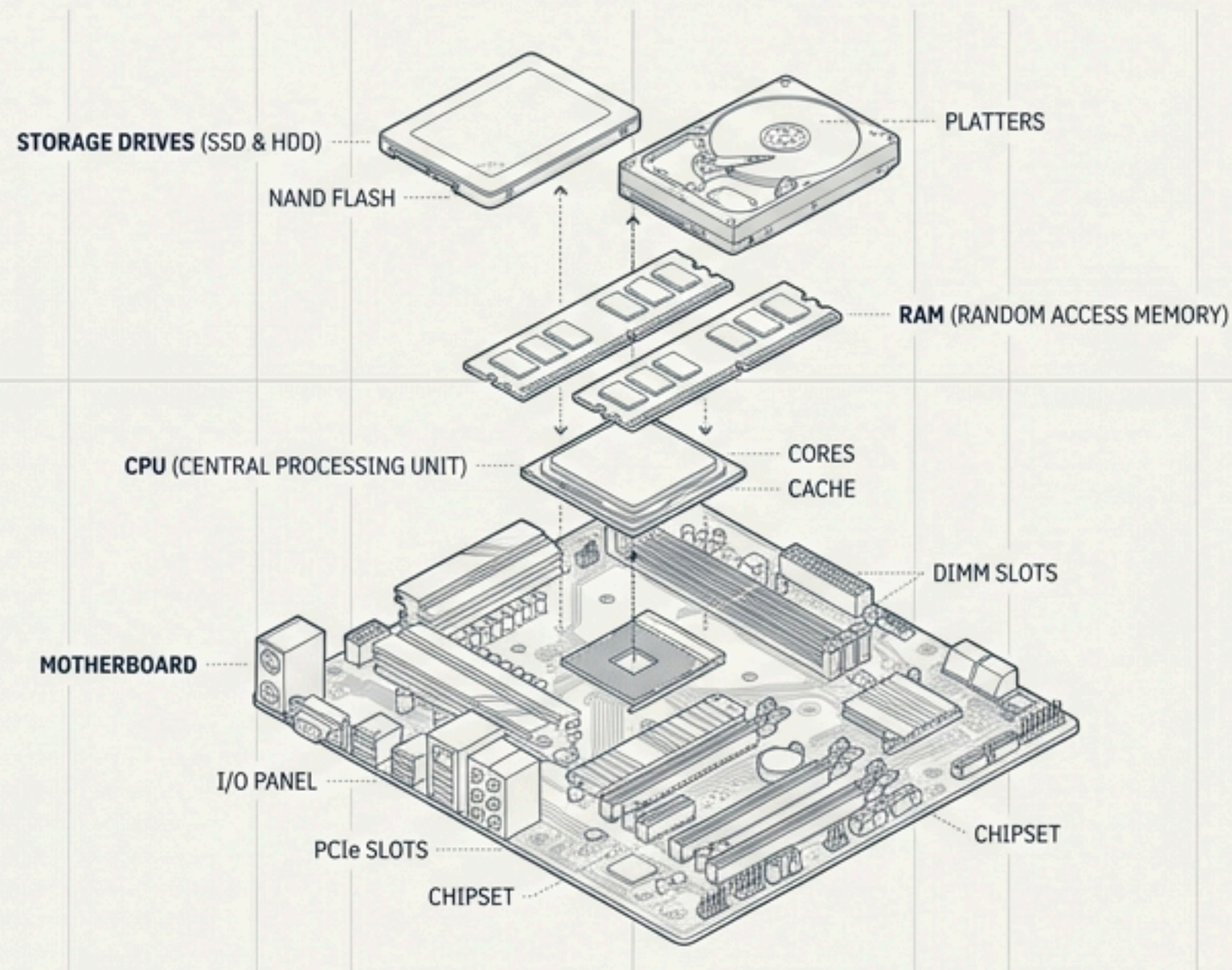


# The Architecture of Intelligence

## COMPUTER SYSTEMS: FROM SILICON ARCHITECTURE TO ABSTRACT DATA

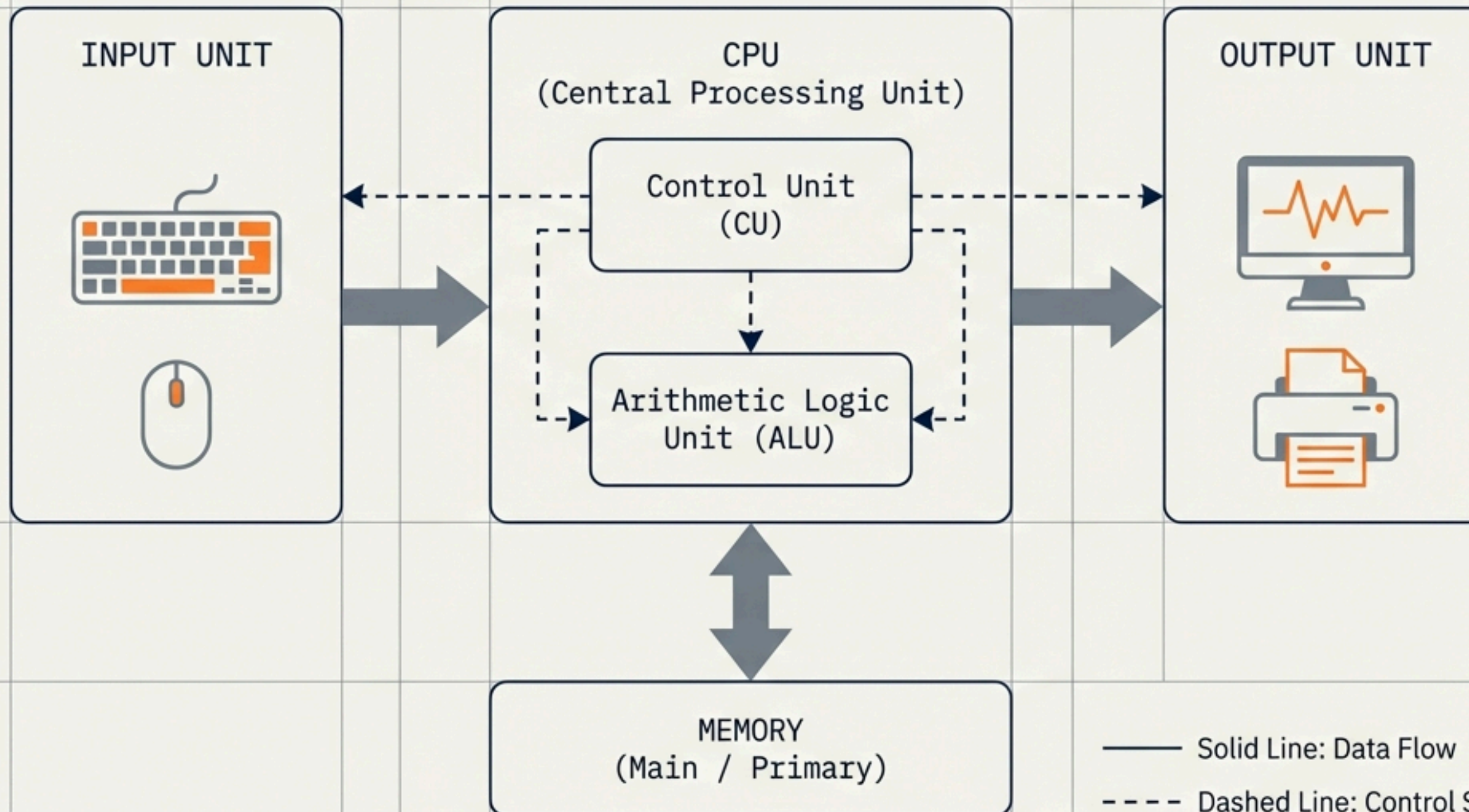


“A computer would deserve to be called intelligent if it could deceive a human into believing that it was human.” — Alan Turing

COMPUTER SYSTEM: A programmable electronic device that accepts data (input), processes it, and generates results (output).



# The Functional Core: Input, Process, Output



**Input:** Converts raw data to digital.

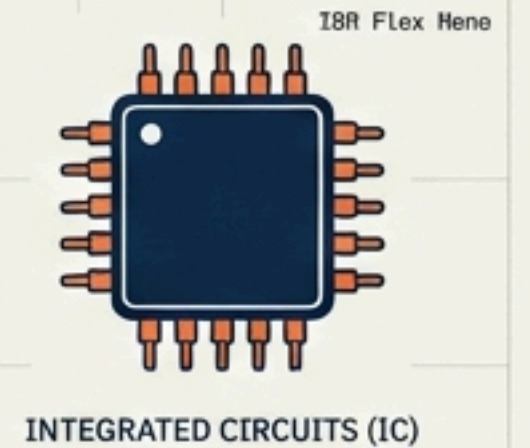
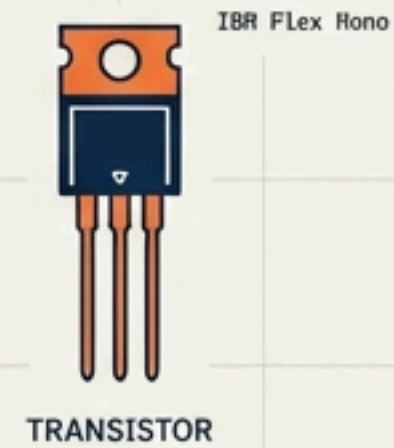
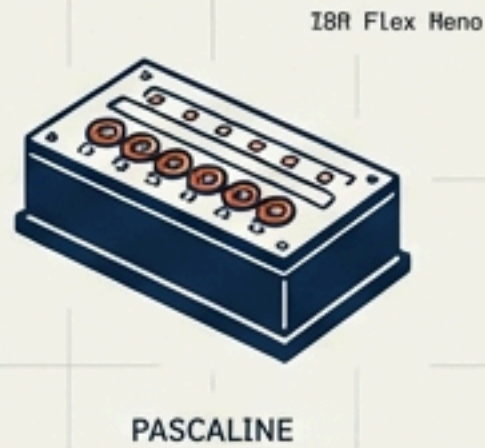
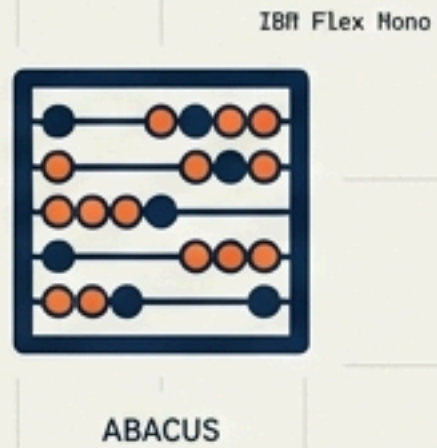
**CPU:** The brain (CU + ALU).

**Memory:** Active workspace.

**Output:** Digital to human-readable.



# The Evolution of Computing Power



3000 BC  
Abacus  
(Mechanical)

1642  
Pascaline  
(Pascal)

1834  
Analytical Engine  
(Babbage)

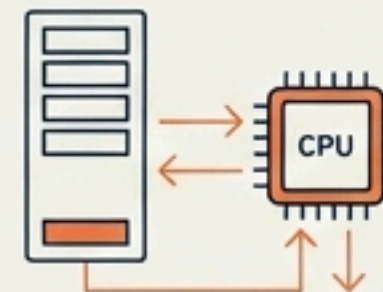
1945  
Von Neumann  
Architecture

1947  
Transistor  
Invention

1970+  
Integrated  
Circuits (IC)

## The Turning Point: Von Neumann Architecture

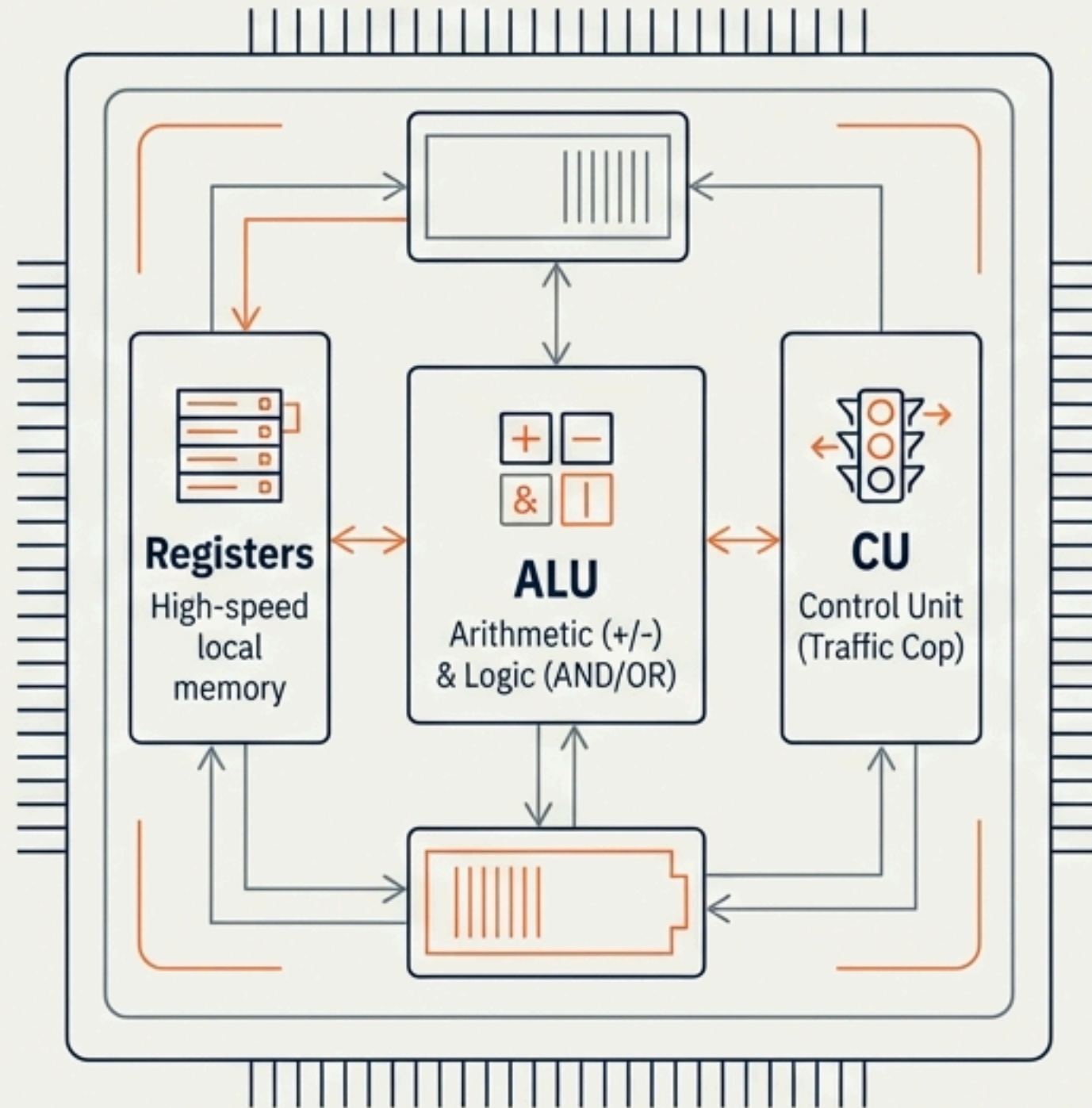
Introduced the 'Stored Program' concept where data and instructions share the same memory space. This fundamental shift enabled general-purpose computing, paving the way for modern processors and the integrated circuits that followed.





# The Brain: CPU and Microprocessors

## Anatomy of the CPU



## Metrics of Power

Clock Speed

**GHz**

GHz (Billions of pulses/sec)



Word Size

**Bits**

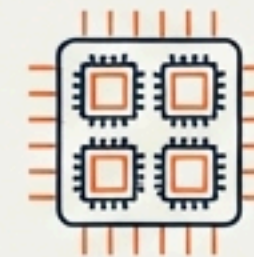
Bits processed simultaneously  
(16 to 64-bit)



Cores

**Units**

Independent processing units  
(Dual, Quad, Octa)

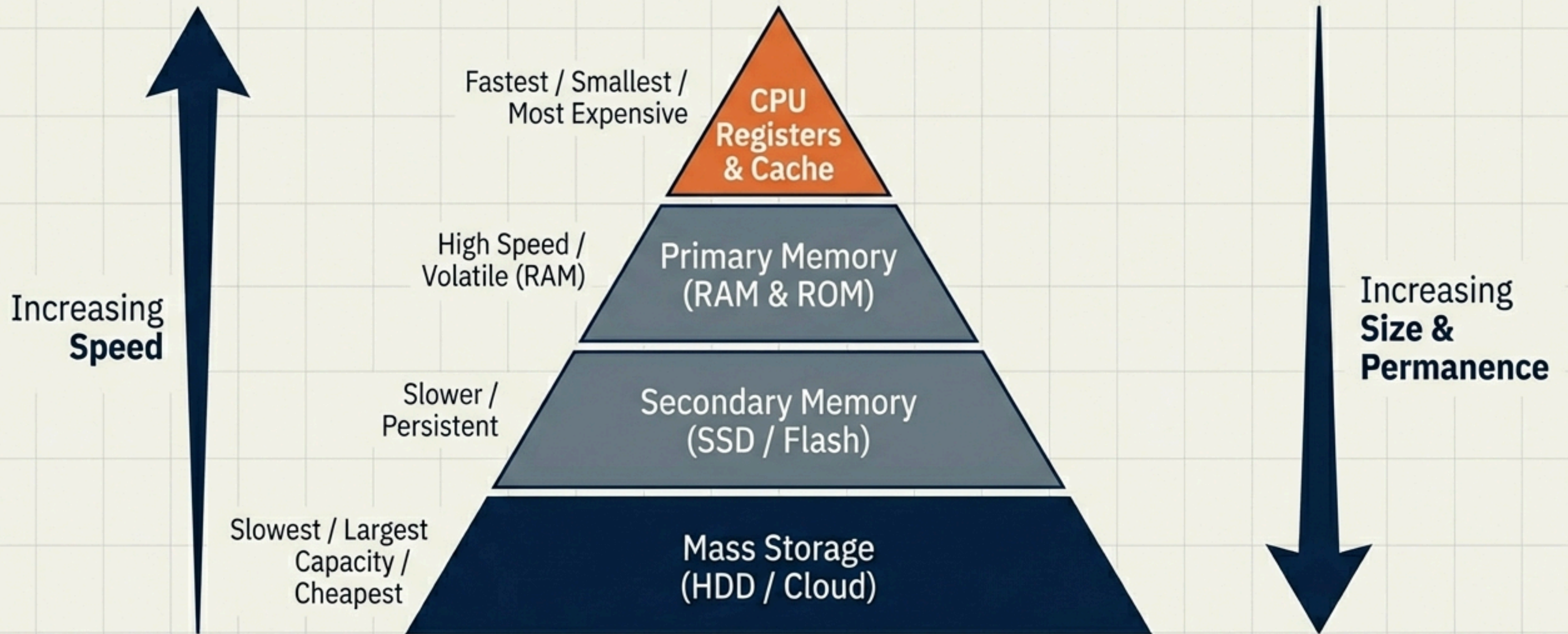


## Moore's Law





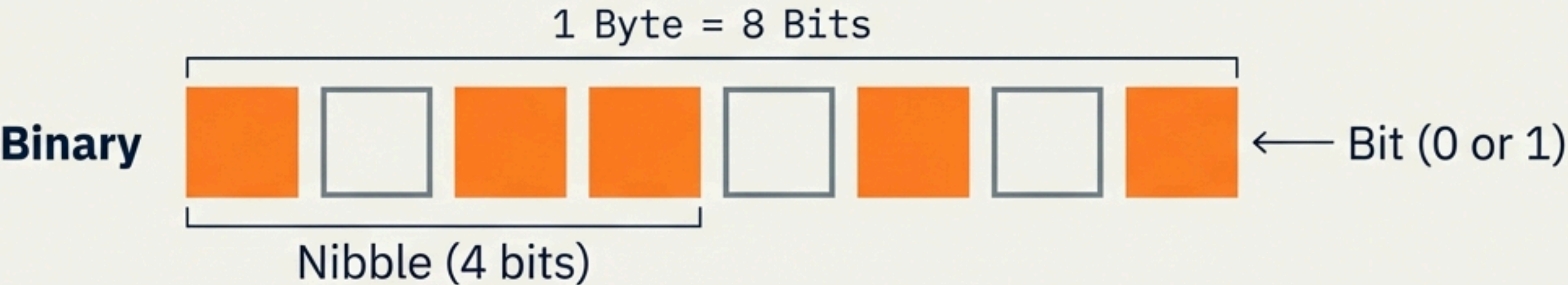
# Memory Hierarchy: Speed vs. Permanence



Primary Memory (RAM) is the active workspace. Secondary Memory is the archive.



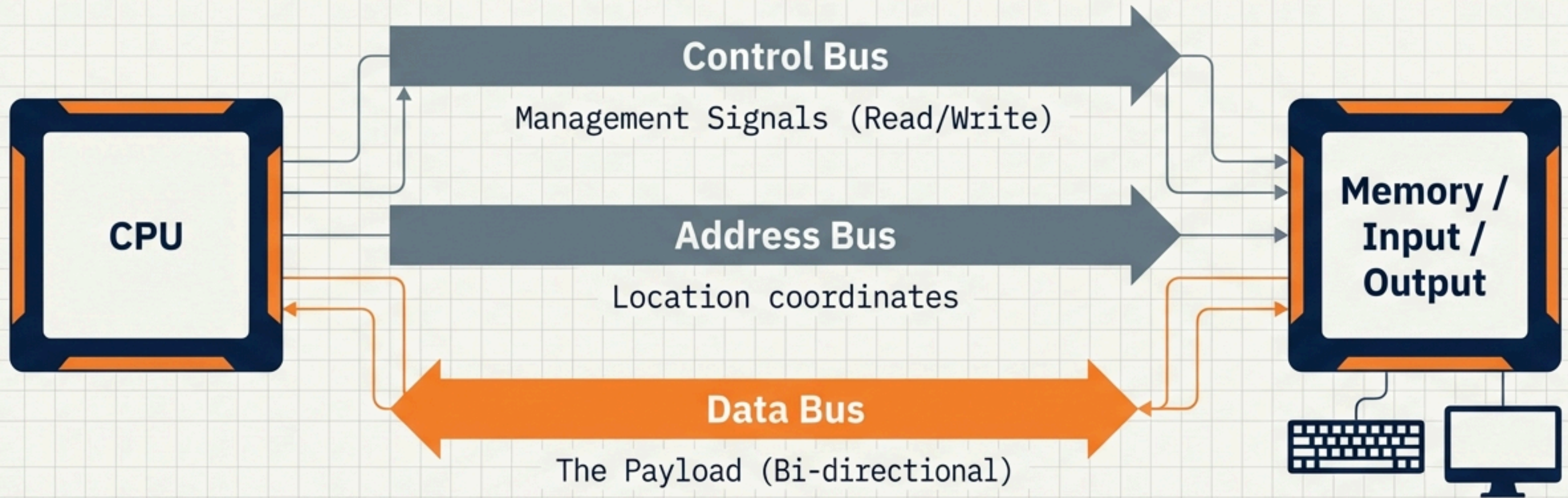
# The Language of Memory



| Unit      | Abbreviation | Equivalent |
|-----------|--------------|------------|
| Kilobyte  | KB           | 1024 Bytes |
| Megabyte  | MB           | 1024 KB    |
| Gigabyte  | GB           | 1024 MB    |
| Terabyte  | TB           | 1024 GB    |
| Petabyte  | PB           | 1024 TB    |
| Exabyte   | EB           | 1024 PB    |
| Zettabyte | ZB           | 1024 EB    |
| Yottabyte | YB           | 1024 ZB    |



# The Nervous System: Data Transfer



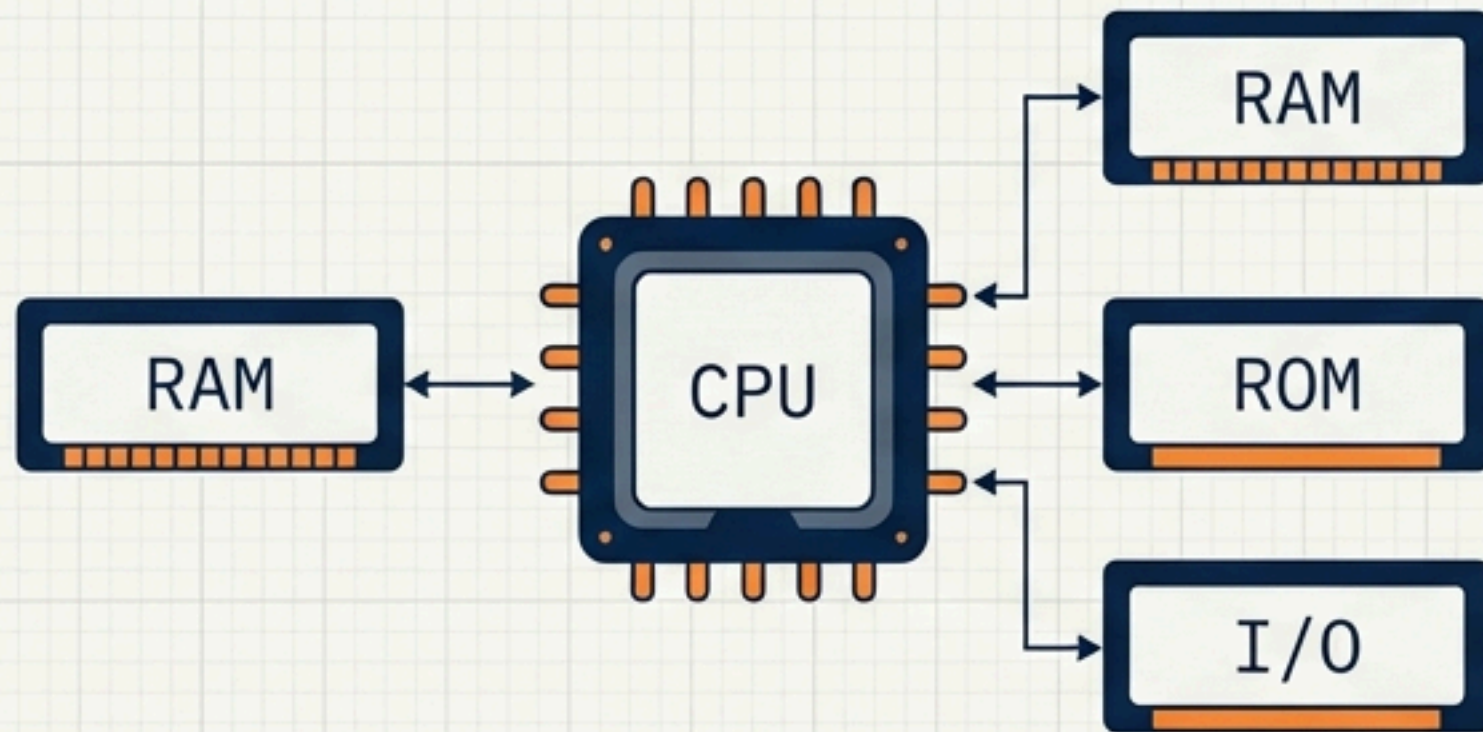
The System Bus connects the brain (CPU) to the body. The CPU sends an address, issues a control signal, and data travels over the Data Bus.



# Generalists vs. Specialists

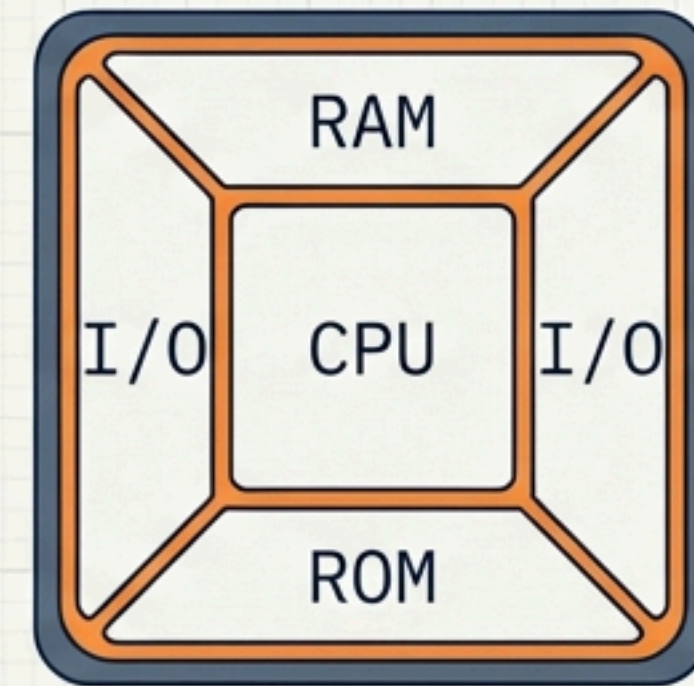


## Microprocessor



The Generalist. Used in PCs.  
High processing power.  
Components are external.

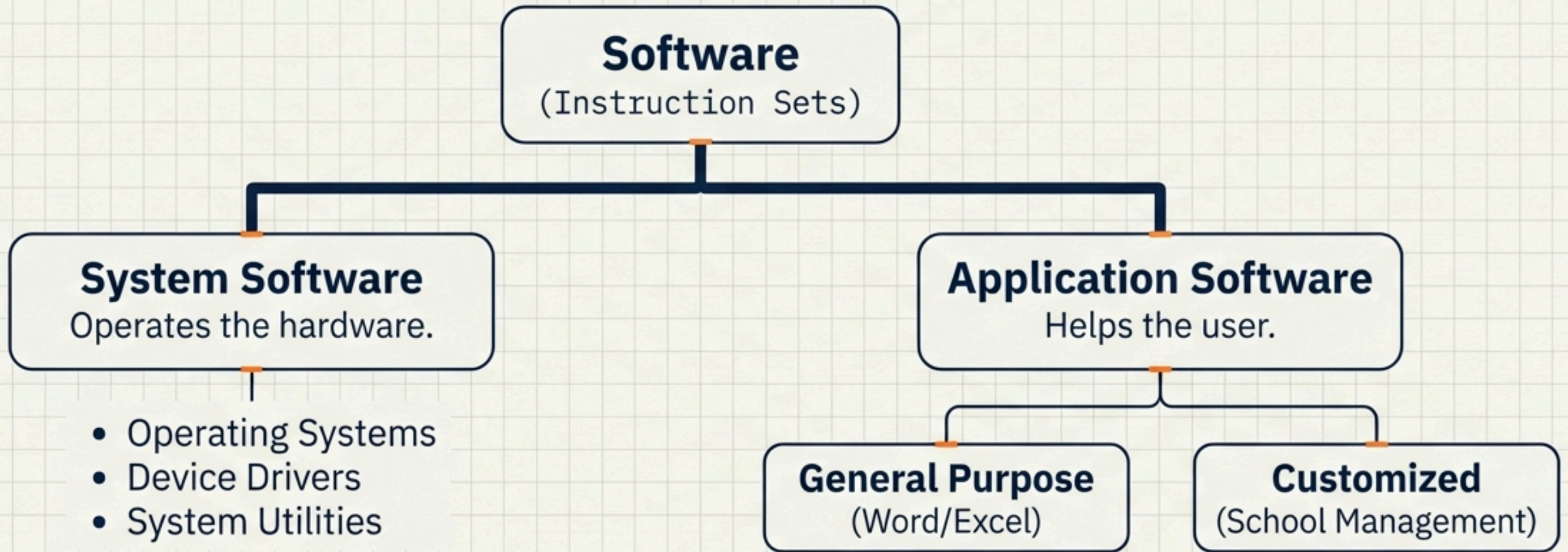
## Microcontroller



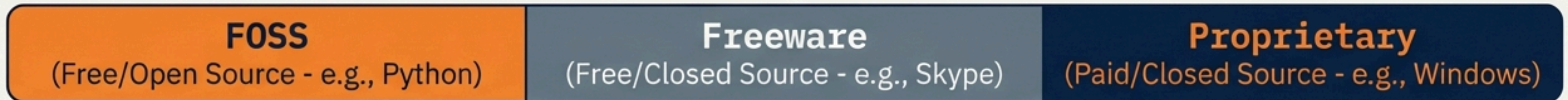
The Specialist. Used in appliances.  
Dedicated task (e.g., wash cycle).  
All-in-one embedded chip.



# The Soul of the Machine: Software

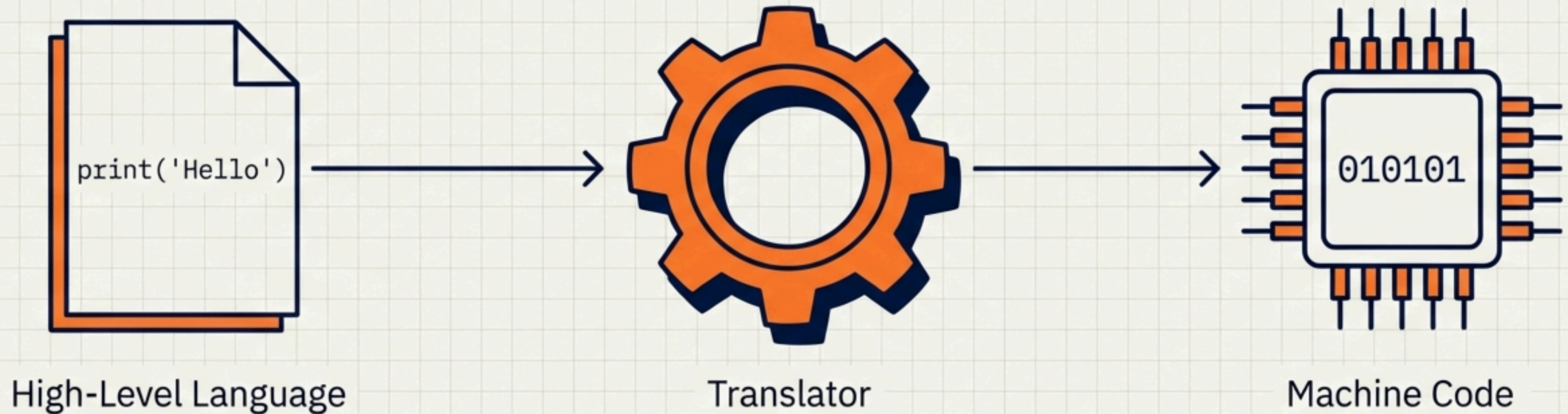


## Ownership Models





# Speaking to the Machine: Programming Tools



## Assembler

Assembly Code ->  
Machine Code.

## Compiler

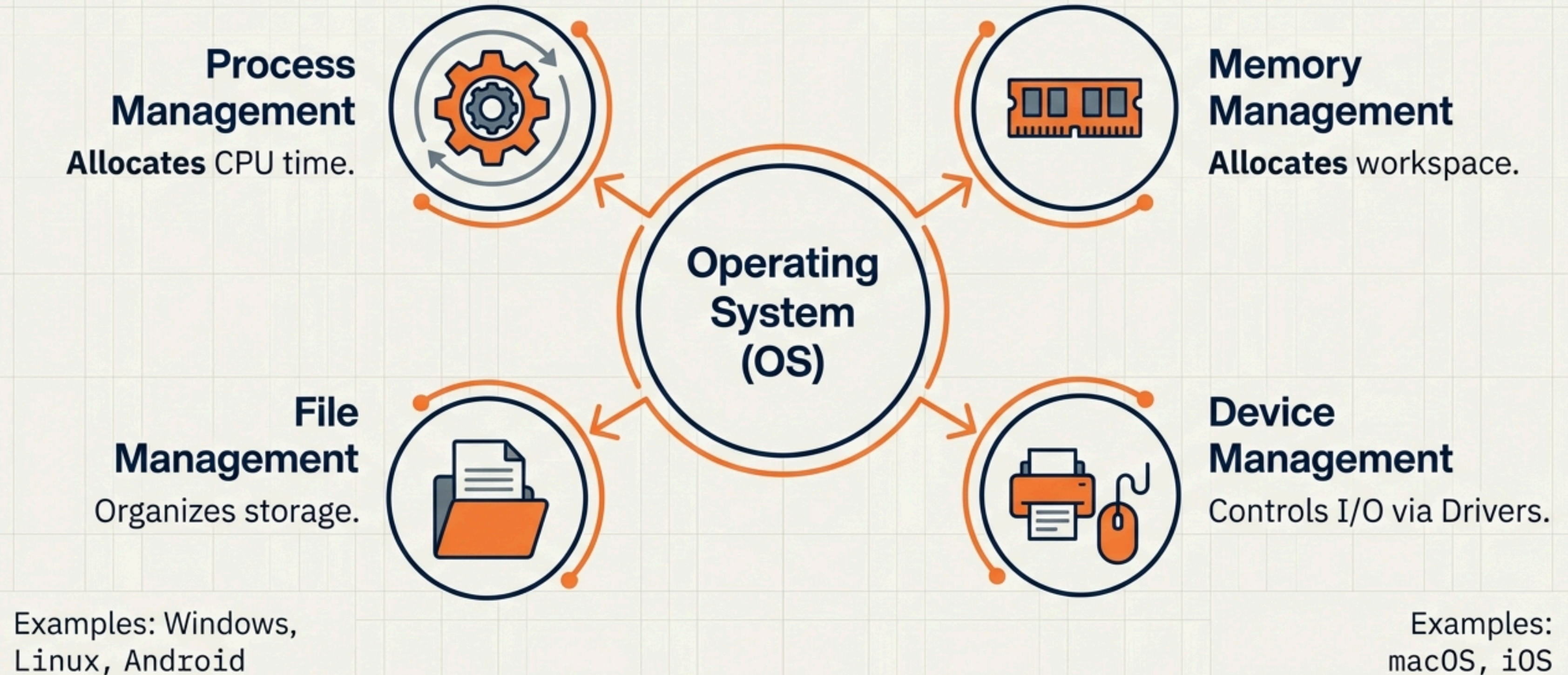
Converts WHOLE code at  
once (Fast execution).

## Interpreter

Converts line-by-line  
(Flexible but slower).

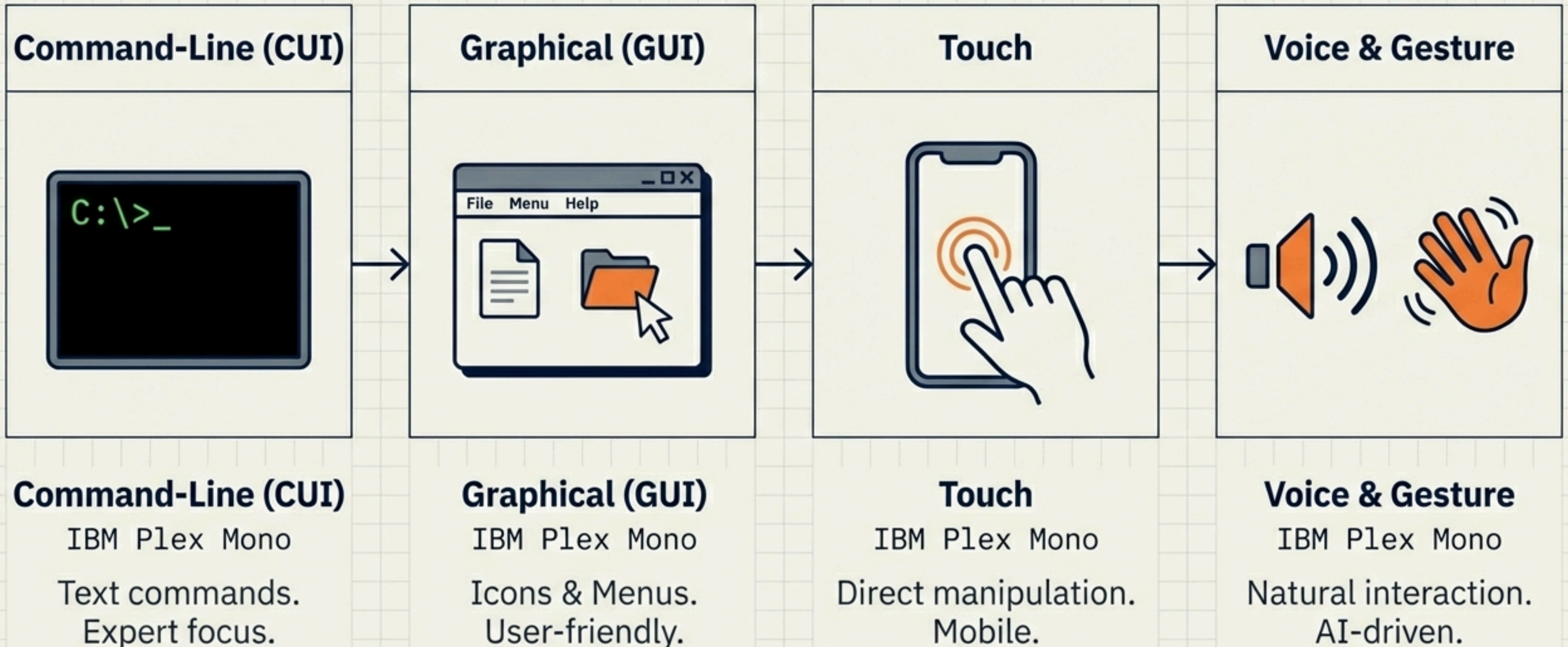


# The Resource Manager: Operating System (OS)





# User Interfaces: Interaction Evolution





# The Raw Material: Data Types

## Structured Data

| ID  | Name     | Date       | Status  |
|-----|----------|------------|---------|
| 001 | A. Smith | 2024-10-27 | Present |
| 002 | B. Jones | 2024-10-27 | Absent  |
|     |          |            |         |
|     |          |            |         |
|     |          |            |         |

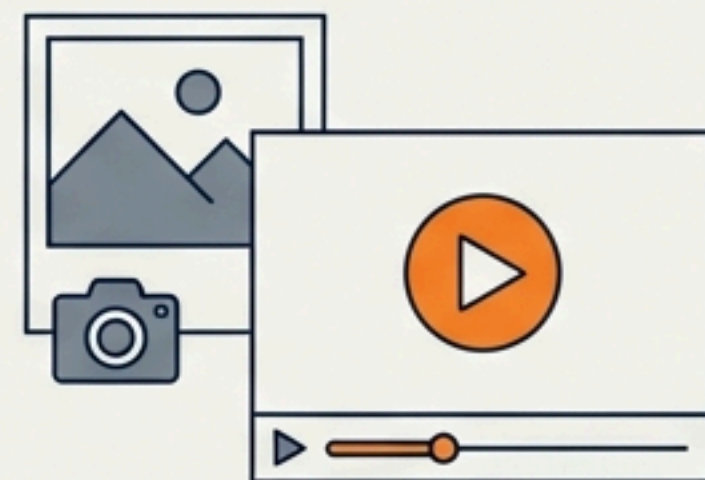
Organized in rows/columns.  
Pre-defined format.  
(e.g., Attendance records).

## Semi-Structured

```
<data>  
  <person>  
    <name>John</name>  
    <role>Engineer</role>  
  </person>  
  <email>john@example.com</email>  
</data>
```

Contains tags but no strict table.  
(e.g., HTML, Emails, CSV).

## Unstructured Data

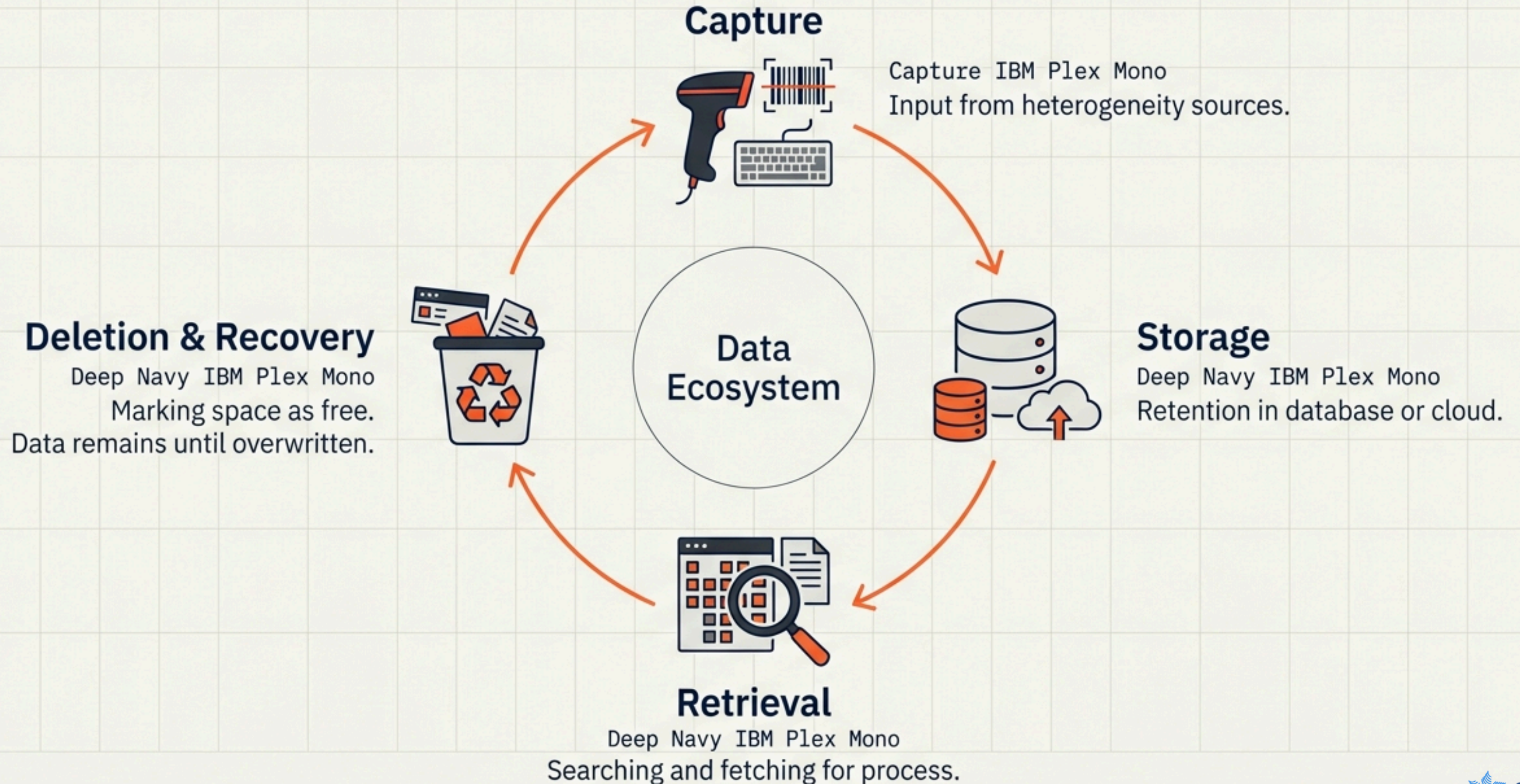


Lorem ipsum  
dolor sit amet,  
consectetur  
adipiscing elit.  
Sed do  
eiusmod  
tempor  
incididunt ut  
labore et  
dolore magna  
aliqua.

No pre-defined format.  
Heavy media. (e.g., Videos,  
Images, Social Posts).

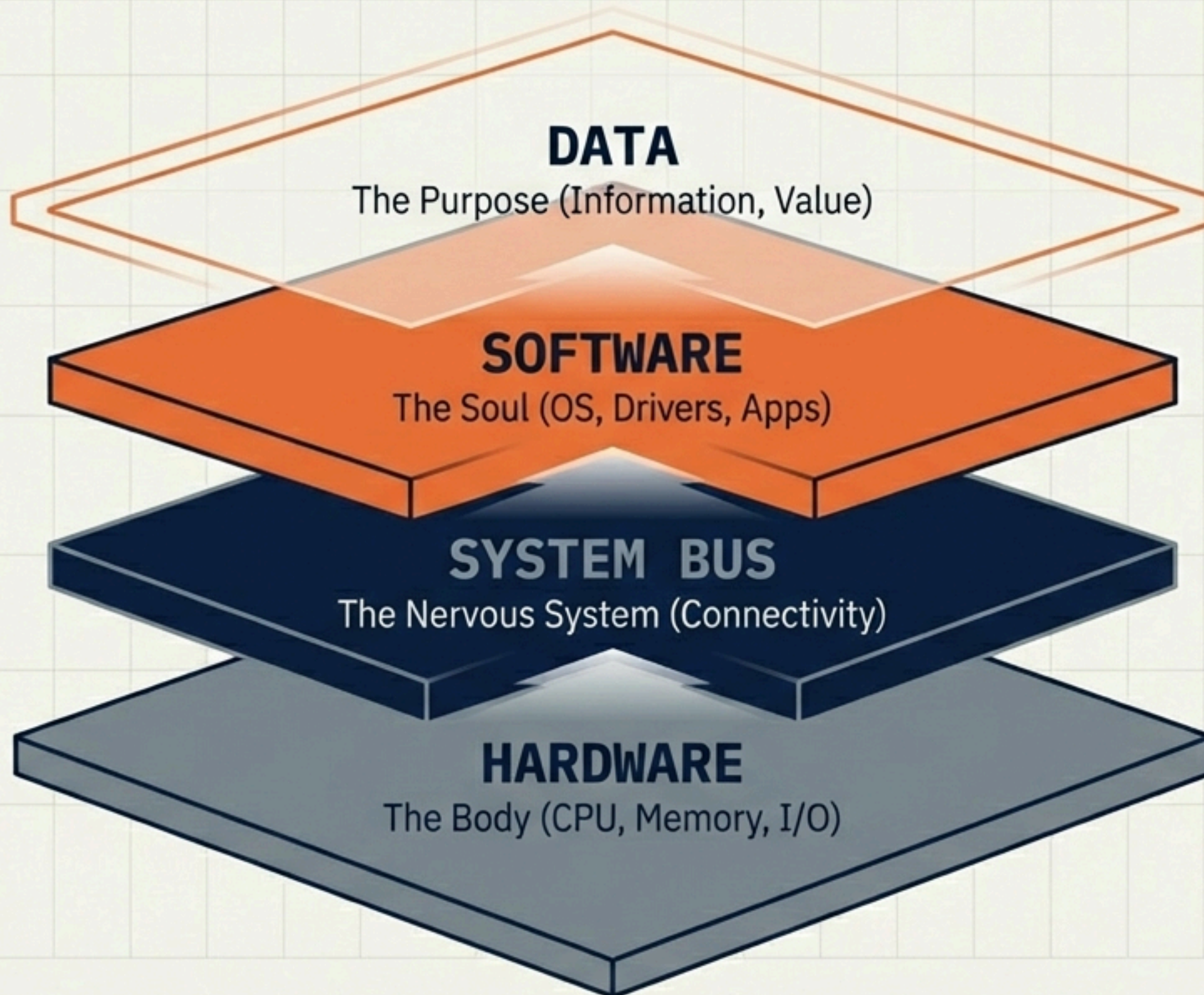


# The Data Lifecycle





# The Convergence: Summary



From the transistor to the file system, the computer system is a layered architecture of intelligence.